

# exercise-bank

Typst Package

*Exercise Management & Banking System*

---

A comprehensive solution for creating, organizing, and filtering exercises

Version 0.6.1

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# 1. Introduction

`exercise-bank` is a Typst package for creating and managing exercises with solutions, metadata, filtering, and exercise banks. Perfect for teachers, textbook authors, and educational content creators.

## 1.1. Features

- Exercises with inline or deferred solutions
- Teacher corrections with flexible display modes
- Difficulty levels shown as badge colors, stars, or symbols
- Clickable links between exercises and their deferred corrections
- Split solution/correction placement, with an inline epigraph-style solution mode
- Chapter-prefixed numbering (“3.5”) and automatic end-of-chapter corrections
- Draft mode for work-in-progress documents
- Multiple display control options (what, where, which content)
- Metadata support (topic, level, author, competencies)
- Exercise banks: define once, display anywhere
- Powerful filtering by any criteria
- Automatic numbering with reset options
- Customizable labels (localization support)

## 1.2. Installation

Import the package in your Typst document:

```
#import "@preview/exercise-bank:0.6.3": exo, exo-setup
```

## 1.3. Quick Start

Code

```
#exo(  
  exercise: [Solve the equation  $2x + 5 = 13$ .]  
)
```

Preview

**Exercise 1** Solve the equation  $2x + 5 = 13$ .

## 2. Basic Usage

### 2.1. Simple Exercise

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.]  
)
```

Preview

**Exercise 1** Calculate  $3 + 4 \times 2$ .

### 2.2. Exercise with Solution

Code

```
#exo(  
  exercise: [Calculate $3 + 4 times 2$.],  
  solution: [$3 + 4 times 2 = 3 + 8 = 11$],  
)
```

Preview

**Exercise 1** Calculate  $3 + 4 \times 2$ .

**Solution 1**  $3 + 4 \times 2 = 3 + 8 = 11$

### 2.3. Multiple Exercises

Exercises are automatically numbered:

Code

```
#exo(exercise: [Simplify $x^2 + 2x + 1$.])  
#exo(exercise: [Factor $x^2 - 4$.])  
#exo(exercise: [Solve $2x - 6 = 0$.])
```

Preview

**Exercise 1** Simplify  $x^2 + 2x + 1$ .

**Exercise 2** Factor  $x^2 - 4$ .

**Exercise 3** Solve  $2x - 6 = 0$ .

### 3. Display Control

The package uses three parameters to control display behavior.

#### 3.1. display - What to Display

Controls what content is displayed:

- "both" (default) - Show exercises and solutions/corrections
- "ex" - Show only exercises
- "sol" - Show only solutions/corrections

Code

```
#exo-setup(display: "ex") // Exercises only

#exo(
  exercise: [Solve  $x + 3 = 7$ .],
  solution: [ $x = 4$ ], // Hidden
)
```

Preview

**Exercise 1** Solve  $x + 3 = 7$ .

#### 3.2. corr-display - Which Content to Show

Controls whether to show solutions or corrections:

- "solution" (default) - Show solution content
- "correction" - Show correction content
- "mixed" - Default to solution, but use correction for exercises with `show-corr: true`

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Simplify  $2x + 3x$ .],
  correction: [
     $2x + 3x = 5x$ 
    (Combine like terms)
  ],
)
```

Preview

**Exercise 1** Simplify  $2x + 3x$ .

**Correction 1**  $2x + 3x = 5x$  (Combine like terms)

#### 3.3. corr-loc - Where to Display

Controls where solutions/corrections appear:

- "after" (default) - Immediately after each exercise
- "pagebreak" - With a page break between
- "end-section" - Collected at section end
- "end-chapter" - Collected at chapter end

## Code

```
#exo-setup(corr-loc: "end-section")

#exo(exercise: [Exercise 1], solution: [Answer 1])
#exo(exercise: [Exercise 2], solution: [Answer 2])

#exo-print-solutions() // Print collected solutions
```

## Preview

**Exercise 1** Exercise 1

**Exercise 2** Exercise 2

### Solutions

**Solution 1** Answer 1

**Solution 2** Answer 2

## 4. Corrections (Teacher Version)

Corrections are detailed solutions for teachers, including pedagogical notes and teaching tips.

### 4.1. Exercise with Correction

Code

```
#exo-setup(corr-display: "correction")

#exo(
  exercise: [Solve  $x^2 = 9$ .],
  correction: [
    *Teacher notes:*
     $x = \pm 3$ 

    Common mistake: forgetting the negative root.
  ],
)
```

Preview

**Exercise 1** Solve  $x^2 = 9$ .

### 4.2. Solutions Only (Answer Key)

Show only solutions for student answer keys:

Code

```
#exo-setup(display: "sol")

#exo(
  exercise: [This is hidden],
  solution: [Only this solution is shown],
)
```

Preview

**Solution 1** Only this solution is shown

### 4.3. The sol-in-corr Flag

When `corr-display: "correction"`, both correction AND solution are shown by default. Use `sol-in-corr: true` on an exercise to indicate the solution is already embedded in the correction (avoiding duplication):



## Code

```
#exo-setup(corr-display: "correction")

// Without sol-in-corr: both correction and solution shown
#exo(
  exercise: [Problem A],
  correction: [Teacher notes only],
  solution: [$x = 5$],
)

// With sol-in-corr: only correction shown
#exo(
  exercise: [Problem B],
  correction: [
    *Solution:* $x = 3$

    _Teaching tip: Watch for sign errors._
  ],
  solution: [$x = 3$],
  sol-in-corr: true, // Solution already in correction
)
```

## Preview

**Exercise 1** Problem A

**Exercise 2** Problem B

## 4.4. Mixed Display Mode

Use `corr-display: "mixed"` to default to solutions while showing corrections for specific exercises:

## Code

```
#exo-setup(corr-display: "mixed")

#exo(
  exercise: [Simple problem],
  solution: [Quick answer],
  correction: [Detailed explanation],
)

#exo(
  exercise: [Complex problem],
  solution: [Answer],
  correction: [Step-by-step solution],
  show-corr: true, // Shows correction
)
```

## Preview

**Exercise 1** Simple problem

**Exercise 2** Complex problem

## 5. Draft Mode

Show placeholders for incomplete exercises during document preparation.

### 5.1. With Draft Mode ON

Code

```
#exo-setup(  
  draft-mode: true,  
  solution-placeholder: [_[To be written]_] ,  
)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - shows placeholder  
)
```

Preview

**Exercise 1** Solve  $x + 5 = 12$

### 5.2. With Draft Mode OFF (Default)

Empty solutions show minimal space without placeholders:

Code

```
#exo-setup(draft-mode: false)  
  
#exo(  
  exercise: [Solve  $x + 5 = 12$ ],  
  solution: [], // Empty - no placeholder  
)
```

Preview

**Exercise 1** Solve  $x + 5 = 12$

## 6. Exercise Banks

Define exercises once, display them anywhere. Perfect for creating reusable exercise collections.

### 6.1. Defining Bank Exercises

Use `exo-define` to register exercises without displaying them:

```
#exo-define(  
  id: "quad-1",  
  exercise: [Solve  $x^2 - 5x + 6 = 0$ .],  
  topic: "quadratics",  
  level: "1M",  
  solution: [ $x = 2$  or  $x = 3$ ],  
)  
  
#exo-define(  
  id: "geom-1",  
  exercise: [Find the area of a circle with radius 5.],  
  topic: "geometry",  
  level: "1M",  
  solution: [ $A = 25\pi$ ],  
)
```

### 6.2. Displaying Bank Exercises

Use `exo-show` to display a specific exercise:

Code

```
#exo-show("quad-1")
```

Preview

**Exercise 1** Solve  $x^2 - 5x + 6 = 0$ .

### 6.3. Selecting Multiple Exercises

Use `exo-select` with filters:

```
// All quadratics exercises  
#exo-select(topic: "quadratics")  
  
// Level 1M exercises only  
#exo-select(level: "1M")  
  
// Multiple topics  
#exo-select(topics: ("quadratics", "geometry"))  
  
// Limit count  
#exo-select(topic: "algebra", max: 5)  
  
// Custom filter  
#exo-select(where: ex => ex.metadata.level == "hard")
```

## 7. Metadata and Filtering

### 7.1. Adding Metadata

Tag exercises for organization and filtering:

```
#exo(  
  exercise: [Solve  $x + 1 = 5$ .],  
  topic: "algebra",  
  level: "easy",  
  authors: ("Prof. Smith",),  
)
```

### 7.2. Filtering Exercises

Display only exercises matching criteria:

```
#exo-filter(topic: "algebra")  
#exo-filter(level: "easy")  
#exo-filter(topic: "algebra", level: "hard")
```

## 8. Competency Tags

Tag exercises with competencies and display them visually.

Code

```
#exo-setup(show-competencies: true)

#exo-define(
  id: "comp-ex",
  exercise: [Solve and explain your reasoning.],
  competencies: ("C1.1", "C2.3", "C4.1"),
  solution: [Detailed solution here],
)

#exo-show("comp-ex")
```

Preview

**Exercise 1** Solve and explain your reasoning.

C1.1 C2.3 C4.1

### 8.1. Filter by Competency

```
#exo-select(competency: "C1.1")
#exo-select(competencies: ("C1.1", "C2.3"))
```

## 9. Configuration

### 9.1. Global Setup

Use `exo-setup` to configure defaults:

```
#exo-setup(  
  // Display control  
  display: "both",           // "ex", "sol", "both"  
  corr-display: "solution",  // "solution", "correction", "mixed"  
  corr-loc: "after",         // "after", "pagebreak", "end-section", "end-chapter"  
  // Labels  
  exercise-label: "Exercise",  
  solution-label: "Solution",  
  correction-label: "Correction",  
  // Counter behavior  
  counter-reset: "section",  // "section", "chapter", "global"  
  // Display options  
  show-id: false,  
  show-competencies: false,  
  // Draft mode  
  draft-mode: false,  
  // Badge styling  
  badge-style: "box",  
  badge-color: black,  
)
```

### 9.2. Badge Styles

Change the visual style of exercise badges:

```
// Circled number (no label text)  
#exo-setup(badge-style: "circled")  
  
// Filled circle with white number  
#exo-setup(badge-style: "filled-circle", badge-color: rgb("#2563eb"))  
  
// Pill-shaped badge  
#exo-setup(badge-style: "pill")  
  
// Arrow tag  
#exo-setup(badge-style: "tag", badge-color: rgb("#1e40af"))
```

Available styles: "box" (default), "circled", "filled-circle", "pill", "tag", "margin", "border-accent", "underline", "rounded-box", "header-card"

### 9.3. Localization

Change labels for different languages:

#### Code

```
// French  
#exo-setup(  
  exercise-label: "Exercice",  
  solution-label: "Solution",  
  correction-label: "Corrigé",  
)
```

#### Preview

**French:** Exercice, Solution, Corrigé  
**German:** Aufgabe, Lösung  
**Spanish:** Ejercicio, Solución

## 9.4. Counter Reset Options

Control when numbering resets:

```
#exo-setup(counter-reset: "section") // Reset each section
#exo-setup(counter-reset: "chapter") // Reset each chapter
#exo-setup(counter-reset: "global")  // Never reset
```

Use hooks to trigger resets:

```
= New Section
#exo-section-start() // Resets if counter-reset: "section"

= New Chapter
#exo-chapter-start() // Resets if counter-reset: "chapter"
```

## 9.5. Advanced Exercises

Mark exercises as advanced to display a visual cue before the label. The default symbol is \*.

Code

```
#exo(
  exercise: [A challenging problem.],
  advanced: true,
)
```

Preview

**\* Exercise 1** A challenging problem.

Customize the symbol:

Code

```
#exo-setup(advanced-symbol: sym.dagger)
#exo(
  exercise: [Advanced with dagger.],
  advanced: true,
)
```

Preview

**† Exercise 1** Advanced with dagger.

Disable with `advanced-symbol: none`.

## 9.6. Optional Exercises

Mark exercises as optional to display the optional-star icon before the label. Students know they can skip these.

Code

```
#exo(
  exercise: [A bonus problem.],
  optional: true,
)
```

Preview

**★ Exercise 1** A bonus problem.

Customize or disable the symbol via `exo-setup`:

```
#exo-setup(optional-symbol: [★]) // custom
#exo-setup(optional-symbol: none) // disable
```

## 9.7. Correction-Given Exercises

The dumbbell icon (corr-given: true) signals that the printed correction will be handed out to students, so they can work independently with the answer key.

### Code

```
#exo(  
  exercise: [Factor  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  corr-given: true,  
)
```

### Preview

✎ Exercise 1 Factor  $x^2 - 9$ .

Customize via `exo-setup(corr-given-symbol: ...)` or disable with `none`.

## 10. QR Codes

Attach a QR code to any exercise (e.g. linking to a video correction or an online resource). Pass a URL string — the code is generated with `tiaoma` — or ready-made content via the `qr` parameter.

### Code

```
#exo(  
  exercise: [Factorise  $x^2 - 9$ .],  
  solution: [ $(x-3)(x+3)$ .],  
  qr: "https://example.com/eq1",  
)
```

### Preview

Exercise 1 Factorise  $x^2 - 9$ .

The QR code is placed automatically based on the active badge style:

- **Badge styles** (box, circled, filled-circle, pill, tag, margin): the QR sits below the badge in the label margin.
- **Full-width styles** (border-accent, underline, rounded-box, header-card): the exercise content wraps around the QR at the top right.

Solution and correction boxes can carry their own separate QR codes via `qr-sol` and `qr-corr`.

### 10.1. Global QR Settings

Control QR appearance globally via `exo-setup`:

### Code

```
#exo-setup(  
  qr-size: 1.5cm,  
  qr-color: rgb("#1e3a8a"),  
  qr-caption: [Correction],  
  show-qr: true,  
)
```

### Preview

Use `show-qr: false` to suppress all QR codes (e.g. for a print version) without removing the URLs from bank definitions.

### 10.2. QR Placement Modes

`qr-position` controls how the exercise body flows around the QR code for full-width badge styles:

| Value            | Behavior                                                                             |
|------------------|--------------------------------------------------------------------------------------|
| "auto" (default) | Placed per badge style: label margin for badge styles, wrapped for full-width styles |



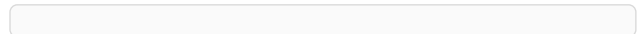
|         |                                                                                                                                                                                                                       |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "wrap"  | Always wrap the exercise text around the QR at the top right, using wrap-it                                                                                                                                           |
| "tasks" | Overlay the QR at the top right without reserving flow height; a taskize <code>#tasks</code> body inside the exercise narrows its own top rows to flow around the QR instead of the whole block being pushed below it |

"tasks" is opt-in and only takes effect when the exercise body actually contains a `#tasks(...)` call from taskize 0.2.8 or newer — the two packages coordinate purely through a shared state key (`taskize-wrap-zone`), with no import dependency in either direction. Bodies without a `#tasks` block ignore the overlay zone and render as if `qr-position` were unset.

#### Code

```
#exo-setup(qr-position: "tasks")
#exo(
  exercise: tasks(
    [First short task.],
    [Second short task.],
    [Third short task.],
  ),
  qr: "https://example.com/quiz",
)
```

#### Preview



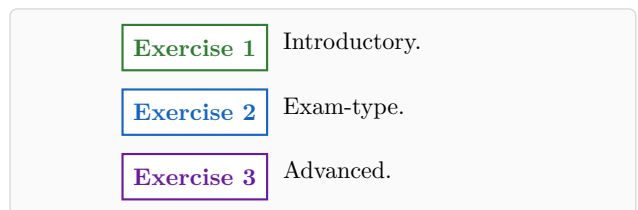
## 11. Difficulty Levels

Tag each exercise with a `difficulty: level` (built-in scale: 1 to 5). By default the exercise badge takes the level color:

#### Code

```
#exo(exercise: [Introductory.],
  difficulty: 1)
#exo(exercise: [Exam-type.],
  difficulty: 3)
#exo(exercise: [Advanced.],
  difficulty: 4)
```

#### Preview



### 11.1. Stars and Symbols

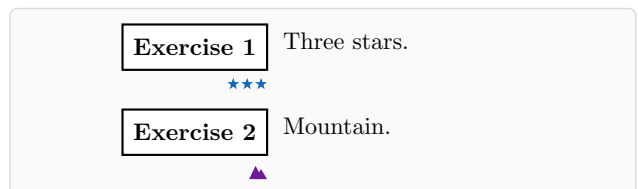
`difficulty-display: "stars"` shows 1–5 small stars; `"symbols"` shows one drawn icon per level (seedling, pencil, target, mountain, star). Both are placed below the badge by default so the badge stays compact (`difficulty-position: "badge"` puts them inline).

#### Code

```
#exo-setup(
  difficulty-display: "stars")
#exo(exercise: [Three stars.],
  difficulty: 3)

#exo-setup(
  difficulty-display: "symbols")
#exo(exercise: [Mountain.],
  difficulty: 4)
```

#### Preview



### 11.2. Custom Scale and Filtering

The scale is a dictionary mapping any key to a color and/or symbol:

```
#exo-setup(difficulty-scale: (
  "easy": (color: rgb("#00897b")),
  "hard": (color: rgb("#e65100"), symbol: [🔥]),
))
#exo(exercise: [...], difficulty: "hard")

// Filtering
#exo-select(difficulty: 3)
#exo-select(difficulties: (1, 2))
#exo-count(difficulty: 4)
```

Difficulty combines well with the optional marker: encode every exercise's level, and mark the mandatory ones with optional: false.

## 12. Solution Placement and Links

### 12.1. Automatic End-of-Chapter Corrections

With corr-loc: "end-chapter" the corrections are only *collected*; print them by calling #exo-chapter-end() where they should appear, or wrap the document with exo-auto-chapter to do it automatically before each new level-1 heading and at the end of the document:

```
#exo-setup(corr-loc: "end-chapter", counter-reset: "chapter")
#show: exo-auto-chapter

= Chapter 1
#exo(exercise: [...], solution: [...])

= Chapter 2 // <- Chapter 1 solutions print just before this title
```

### 12.2. Separate Solution and Correction Placement

sol-loc controls where *solutions* go, independently of corr-loc (default auto = follow it). Typical setup: short answer under the statement, full correction at the end of the chapter:

```
#exo-setup(
  corr-display: "correction", // show correction and solution
  sol-loc: "after",           // answer below the statement
  corr-loc: "end-chapter",    // correction at chapter end
  solution-style: "inline",   // epigraph-like, no badge
)
```

### 12.3. Inline Solution Style

solution-style: "inline" renders the solution as a short rule + content right under the statement, with a small margin label (customizable via inline-label, none to hide; rule length via inline-rule-length):

Code

```
#exo-setup(
  solution-style: "inline")
#exo(
  exercise: [Solve
     $x^2 - 5x + 6 = 0$ .],
  solution: [ $x$  in {2, 3}],
)
```

Preview

|                   |                            |
|-------------------|----------------------------|
| <b>Exercise 1</b> | Solve $x^2 - 5x + 6 = 0$ . |
| <i>Solution</i>   | $x \in \{2, 3\}$           |

## 12.4. Clickable Exercise ↔ Correction Links

When corrections are deferred, `link-solutions: true` adds a clickable arrow icon next to the exercise badge that jumps to the correction, and a back-link on the correction. With `link-style: "page"` the exercise instead shows a textbook-style clickable reference (“Solution p. 30”) at the top right of the statement:

```
#exo-setup(corr-loc: "end-chapter", link-solutions: true)
// or, textbook style:
#exo-setup(corr-loc: "end-chapter", link-solutions: true,
  link-style: "page")
```

Customize with `link-icon / backlink-icon` (icons) or `page-ref-color / page-ref-format` (page reference).

## 12.5. Chapter-Prefixed Numbering

`number-prefix: "heading"` prefixes displayed numbers with the current level-1 heading number, e.g. exercise 5 of chapter 3 shows as “3.5” (separator via `number-separator`):

```
#set heading(numbering: "1.")
#exo-setup(number-prefix: "heading", counter-reset: "chapter")
#show: exo-auto-chapter
```

`number-prefix` also accepts a counter or a function `() => value`, for heading packages that keep their own chapter counter instead of `counter(heading)`. With `beautitled 0.3.0+` the native counter is kept in sync, so `number-prefix: "heading"` works directly; for earlier versions (or with `enable-parts`, where the first heading level is the part) use `beautitled`’s exported counter:

```
#import "@preview/beautitled:0.2.7": beautitled-init, chapter-counter
#show: beautitled-init
#exo-setup(number-prefix: chapter-counter, counter-reset: "chapter")
#show: exo-auto-chapter
```

With `beautitled`’s direct `#chapter(...)` calls (no native headings), wrap the call instead of using `exo-auto-chapter`:

```
#let chapitre(..args) = {
  exo-chapter-end(); chapter(..args); exo-chapter-start()
}
```

## 13. Visual Styles

Set the badge style with `exo-setup(badge-style: "...")`.

### 13.1. Box (Default)

Code

```
#exo-setup(badge-style: "box")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

**Exercise 1** Solve  $x + 3 = 7$

### 13.2. Circled

Code

```
#exo-setup(badge-style: "circled")
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

**1** Solve  $x + 3 = 7$

### 13.3. Filled Circle

Code

```
#exo-setup(  
  badge-style: "filled-circle",  
  badge-color: rgb("#2563eb"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

**1** Solve  $x + 3 = 7$

### 13.4. Rect and Filled Rect

Compact number-only rectangles – a minimal alternative when circles look too large in your font:

Code

```
#exo-setup(badge-style: "rect")  
#exo(exercise: [Solve  $x + 3 = 7$ ])  
#exo-setup(  
  badge-style: "filled-rect",  
  badge-color: rgb("#1a4d8f"),  
)  
#exo(exercise: [Solve  $2x = 10$ ])
```

Preview

**1** Solve  $x + 3 = 7$   
**2** Solve  $2x = 10$

### 13.5. Custom Badge Function

For full control, pass a function (label, number, font-size, color, is-solution) => content as badge-style:

Code

```
#exo-setup(badge-style:  
  (label, number, font-size,  
   color, is-solution) => {  
    box(stroke:  
      (bottom: 1.5pt + color),  
      inset: (x: 4pt, y: 3pt),  
      text(weight: "bold",  
        size: font-size,  
        fill: color)[#number.])  
  })  
#exo(exercise: [Solve  $x^2 = 4$ ])
```

Preview

1. Solve  $x^2 = 4$

### 13.6. Pill

Code

```
#exo-setup(badge-style: "pill")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve  $x + 3 = 7$

### 13.7. Tag

Code

```
#exo-setup(  
  badge-style: "tag",  
  badge-color: rgb("#1e40af"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1 Solve  $x + 3 = 7$

## 13.8. Border Accent

Code

```
#exo-setup(  
  badge-style: "border-accent",  
  badge-color: rgb("#3b82f6"),  
)  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve  $x + 3 = 7$

## 13.9. Underline

Code

```
#exo-setup(badge-style: "underline")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve  $x + 3 = 7$

## 13.10. Rounded Box

Code

```
#exo-setup(badge-style: "rounded-box")  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve  $x + 3 = 7$

## 13.11. Header Card

Code

```
#exo-setup(badge-style: "header-card", badge-color: rgb("#3b82f6"))  
#exo(exercise: [Solve  $x + 3 = 7$ ])
```

Preview

Exercise 1

Solve  $x + 3 = 7$

## 13.12. Margin

The label appears flush-right in a fixed-width left column; content flows in the right column. Solutions are shown in a compact bordered block.

## Code

```
#exo-setup(badge-style: "margin")
#exo(
  exercise: [Solve  $x + 3 = 7$ ],
  solution: [ $x = 4$ ],
)
```

## Preview

**Exercise 1:** Solve  $x + 3 = 7$

**Solution 1**

$$x = 4$$

## 13.13. Badge Position

Every style above places the badge in its own column on the left, so the statement is indented for the whole height of the exercise. In a narrow measure – two-column layouts especially – that column costs its width on **every** line, and it gets worse when the statement holds an enumeration, whose own indent stacks on top of it.

`badge-position: "above"` puts the badge alone on a header line and lets the statement run the full width underneath. It is independent of `badge-style`, so every badge look keeps working:

## Code

```
#exo-setup(badge-style: "filled-circle")
#exo(exercise: [...])           // badge-position: "margin" (default)

#exo-setup(badge-position: "above")
#exo(exercise: [...])           // same badge, full-width statement
```

## Preview

*badge-position: "margin"*

- 1 A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.
- Express the side length of the first tiles.
  - How many tiles of side  $x + 5$  would be needed?

*badge-position: "above"*

- 2 A floor is covered with 500 square tiles. Tiles 5 cm longer and wider would have taken 320 of them.
- Express the side length of the first tiles.
  - How many tiles of side  $x + 5$  would be needed?

With `link-style: "page"` the “Solution p. 34” reference moves onto the badge line as well, instead of being wrapped into the top right of the statement.

## 14. Parameter Reference

### 14.1. `exo` Function

| Parameter                   | Type           | Default | Description                                                  |
|-----------------------------|----------------|---------|--------------------------------------------------------------|
| <code>exercise</code>       | content        | none    | Exercise content (required)                                  |
| <code>solution</code>       | content        | none    | Solution content                                             |
| <code>correction</code>     | content        | none    | Correction for teachers                                      |
| <code>id</code>             | string         | auto    | Unique exercise ID                                           |
| <code>margin-content</code> | content        | none    | Content placed below the badge (e.g. remarks)                |
| <code>qr</code>             | string/content | none    | QR code for the exercise box (URL string or content)         |
| <code>qr-sol</code>         | string/content | none    | QR code for the solution box                                 |
| <code>qr-corr</code>        | string/content | none    | QR code for the correction box                               |
| <code>sol-in-corr</code>    | bool           | false   | If true, solution is already in correction (don't show both) |
| <code>show-corr</code>      | bool           | false   | If true, show correction in “mixed” mode                     |
| <code>optional</code>       | bool           | false   | Show the optional marker before the label                    |
| <code>corr-given</code>     | bool           | false   | Show the correction-given marker (dumbbell icon)             |
| <code>topic</code>          | string         | none    | Topic metadata                                               |
| <code>level</code>          | string         | none    | Difficulty level                                             |
| <code>authors</code>        | array          | ()      | Author names                                                 |

### 14.2. `exo-define` Function

Same as `exo`, plus:

| Parameter                 | Type   | Default | Description                                      |
|---------------------------|--------|---------|--------------------------------------------------|
| <code>optional</code>     | bool   | false   | Show the optional marker before the label        |
| <code>corr-given</code>   | bool   | false   | Show the correction-given marker (dumbbell icon) |
| <code>competencies</code> | array  | ()      | Competency tags                                  |
| <code>points</code>       | number | none    | Points (for exam mode)                           |

### 14.3. `exo-select` Function

| Parameter               | Type   | Default | Description          |
|-------------------------|--------|---------|----------------------|
| <code>topic</code>      | string | none    | Filter by topic      |
| <code>level</code>      | string | none    | Filter by level      |
| <code>author</code>     | string | none    | Filter by author     |
| <code>competency</code> | string | none    | Filter by competency |
| <code>topics</code>     | array  | none    | Filter by any topic  |

|              |          |      |                          |
|--------------|----------|------|--------------------------|
| levels       | array    | none | Filter by any level      |
| competencies | array    | none | Filter by any competency |
| where        | function | none | Custom filter function   |
| renumber     | bool     | true | Renumber sequentially    |
| max          | int      | none | Maximum exercises        |

## 14.4. **exo-setup** Function

### Display Control:

| Parameter              | Type                              | Default                | Description                                                         |
|------------------------|-----------------------------------|------------------------|---------------------------------------------------------------------|
| display                | string                            | “both”                 | “ex”, “sol”, “both”                                                 |
| corr-display           | string                            | “solution”             | “solution”, “correction”, “mixed”                                   |
| corr-loc               | string                            | “after”                | “after”, “pagebreak”, “end-section”, “end-chapter”                  |
| sol-loc                | string/auto                       | auto                   | Same values as <code>corr-loc</code> , for solutions only           |
| exercise-label         | string                            | “Exercise”             | Label for exercises                                                 |
| solution-label         | string                            | “Solution”             | Label for solutions                                                 |
| correction-label       | string                            | “Correction”           | Label for corrections                                               |
| counter-reset          | string                            | “section”              | “section”, “chapter”, “global”                                      |
| number-prefix          | none/str/<br>counter/<br>function | none                   | “heading” (level-1 heading number), a custom counter, or a function |
| number-separator       | string                            | “.”                    | Separator for chapter-prefixed numbers                              |
| show-id                | bool                              | false                  | Show exercise IDs                                                   |
| show-competencies      | bool                              | false                  | Show competency tags                                                |
| draft-mode             | bool                              | false                  | Show placeholders for empty content                                 |
| solution-placeholder   | content                           | <i>To be completed</i> | Placeholder text                                                    |
| correction-placeholder | content                           | <i>To be completed</i> | Placeholder text                                                    |

### Visual Styling:

| Parameter      | Type            | Default  | Description                                                                                                                                                                       |
|----------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| badge-style    | string/function | “box”    | Style: “box”, “circled”, “filled-circle”, “rect”, “filled-rect”, “pill”, “tag”, “margin”, “border-accent”, “underline”, “rounded-box”, “header-card” – or a custom badge function |
| badge-position | string          | “margin” | “margin” (badge in its own left column) or “above” (badge on a header line, statement full width below)                                                                           |
| badge-color    | color           | black    | Color for exercise badges                                                                                                                                                         |



|                  |             |       |                                  |
|------------------|-------------|-------|----------------------------------|
| solution-color   | color       | green | Color for solution badges        |
| correction-color | color       | green | Color for correction badges      |
| label-font-size  | length      | 12pt  | Font size for badge labels       |
| margin-position  | length/auto | auto  | Width reserved for badge column  |
| label-extra      | length      | 1cm   | Extra space for labels in margin |

### Spacing:

| Parameter         | Type         | Default       | Description                                       |
|-------------------|--------------|---------------|---------------------------------------------------|
| exercise-above    | length       | 0.8em         | Space above exercise boxes                        |
| exercise-below    | length       | 0.8em         | Space below exercise boxes                        |
| solution-above    | length       | 0.8em         | Space above solution boxes                        |
| solution-below    | length       | 0.8em         | Space below solution boxes                        |
| correction-above  | length       | 0.8em         | Space above correction boxes                      |
| correction-below  | length       | 0.8em         | Space below correction boxes                      |
| advanced-symbol   | content/none | "*"           | Symbol before label for advanced exercises        |
| optional-symbol   | content/none | star icon     | Symbol before label for optional exercises        |
| corr-given-symbol | content/none | dumbbell icon | Symbol before label when correction is handed out |

### Difficulty, Inline Solutions, and Links:

| Parameter           | Type              | Default    | Description                                                           |
|---------------------|-------------------|------------|-----------------------------------------------------------------------|
| difficulty-display  | string            | "color"    | "color", "stars", "symbols", "none"                                   |
| difficulty-scale    | auto/dict         | auto       | auto = built-in 5-level scale, or dict key -> (color: .., symbol: ..) |
| difficulty-position | string            | "below"    | Stars/symbols "below" the badge or inline in the "badge"              |
| solution-style      | auto/string       | auto       | "inline" = epigraph-style rule instead of a badge box                 |
| inline-rule-length  | length            | 3cm        | Rule length for inline solutions                                      |
| inline-label        | auto/content/none | auto       | Small margin label for inline solutions                               |
| link-solutions      | bool              | false      | Clickable links to deferred corrections                               |
| link-style          | string            | "icon"     | "icon" (arrow) or "page" ("Solution p. 30" reference)                 |
| link-icon           | content/none      | arrow icon | Link icon on the exercise                                             |
| backlink-icon       | content/none      | arrow icon | Back-link icon on the correction                                      |
| page-ref-format     | auto/function     | auto       | Custom page reference: (label, page) => content                       |

|                |            |      |                                           |
|----------------|------------|------|-------------------------------------------|
| page-ref-color | auto/color | auto | Page reference color (auto = badge color) |
|----------------|------------|------|-------------------------------------------|

## QR Code Settings:

| Parameter   | Type         | Default | Description                                                                                                                |
|-------------|--------------|---------|----------------------------------------------------------------------------------------------------------------------------|
| show-qr     | bool         | true    | Master toggle for per-exercise QR codes                                                                                    |
| qr-size     | length       | 1.5cm   | Target QR code size (shrinks to fit margin if needed)                                                                      |
| qr-min-size | length       | 1cm     | Minimum size; below this the QR extends into page margin                                                                   |
| qr-color    | color        | black   | QR module color                                                                                                            |
| qr-caption  | content/none | none    | Small caption below every QR code                                                                                          |
| qr-position | string       | “auto”  | “auto” (per badge style), “wrap” (always wrap content), or “tasks” (overlay; a taskize <b>#tasks</b> body flows around it) |

## 15. Utility Functions

| Function                             | Description                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <code>exo-reset-counter()</code>     | Reset exercise numbering to 0                                                                          |
| <code>exo-clear-registry()</code>    | Clear all registered exercises                                                                         |
| <code>exo-get-registry()</code>      | Retrieve the current registry of exercise metadata (context-dependent)                                 |
| <code>exo-section-start()</code>     | Call at section start (triggers reset if configured)                                                   |
| <code>exo-chapter-start()</code>     | Call at chapter start (triggers reset if configured)                                                   |
| <code>exo-section-end()</code>       | Print solutions pending for end-of-section placement                                                   |
| <code>exo-chapter-end()</code>       | Print solutions pending for end-of-chapter placement                                                   |
| <code>exo-print-solutions()</code>   | Print collected solutions (for end-section/chapter modes)                                              |
| <code>exo-count(topic: ..)</code>    | Count exercises matching criteria                                                                      |
| <code>exo-show("id")</code>          | Display exercise by ID                                                                                 |
| <code>exo-show-many("a", "b")</code> | Display multiple exercises by ID                                                                       |
| <code>exo-filter(topic: ..)</code>   | Filter and display matching exercises                                                                  |
| <code>competency-tag("comp")</code>  | Render a single competency as a small styled pill (used internally by <code>show-competencies</code> ) |

## 16. Exam Integration

These functions support building exam documents from an exercise bank, typically used alongside a layout package such as `texam`.

### 16.1. exam-setup

Configure the exam correction box appearance and behaviour.

| Parameter                         | Type   | Default      | Description                                        |
|-----------------------------------|--------|--------------|----------------------------------------------------|
| <code>show-solutions</code>       | bool   | false        | Show <code>exam-solution-box</code> content        |
| <code>solution-label</code>       | string | "Correction" | Label shown in bold at the top of the solution box |
| <code>solution-fill</code>        | color  | green tint   | Background fill of the solution box                |
| <code>solution-stroke</code>      | stroke | green stroke | Border stroke of the solution box                  |
| <code>solution-radius</code>      | length | 3pt          | Corner radius of the solution box                  |
| <code>solution-inset</code>       | length | 8pt          | Inner padding of the solution box                  |
| <code>solution-label-color</code> | color  | dark green   | Color of the label text                            |
| <code>question-spacing</code>     | length | 1em          | Vertical space between questions                   |

### 16.2. exam-solution-box

A green-highlighted answer block. Only rendered when `exam-setup(show-solutions: true)`.

```
#exam-solution-box[
  $(x - 2)(x - 3)$
]
```

Place it after an exercise. The same source file produces both the student version (box hidden) and the correction (box shown) by flipping `show-solutions`.

### 16.3. exam-score-table

Builds a score grid from registered exercise IDs, reading point values from the registry.

```
#exam-score-table("ex-1", "ex-2", "ex-3")
```

| Parameter      | Type    | Default    | Description                              |
|----------------|---------|------------|------------------------------------------|
| ..exercise-ids | strings |            | Positional — IDs of exercises to include |
| question-label | string  | "Question" | Header for the question column           |
| points-label   | string  | "Points"   | Header for the points row                |
| grade-label    | string  | "Note"     | Header for the grade row                 |
| total-label    | string  | "Total"    | Label for the total column               |